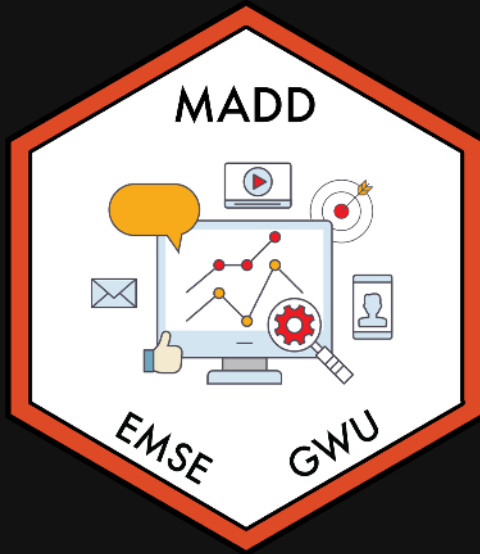




Week 8: *Optimization & MLE*

 EMSE 6035: Marketing Analytics for Design Decisions

 John Paul Helveston

 October 15, 2025

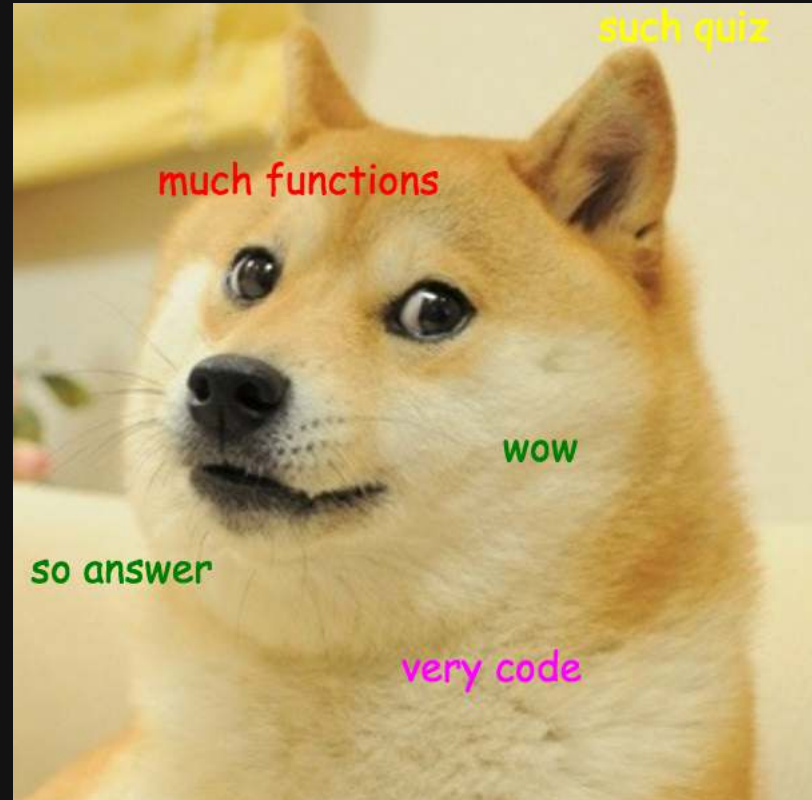
Quiz 3

Download the template from the #class channel

Make sure you unzip it!

When done, submit your **quiz3.qmd** on Blackboard

10:00



Week 8: *Optimization & MLE*

1. Maximum likelihood estimation

2. Optimization (in general)

BREAK

3. Joins

4. Pilot data cleaning

Week 8: *Optimization & MLE*

1. Maximum likelihood estimation

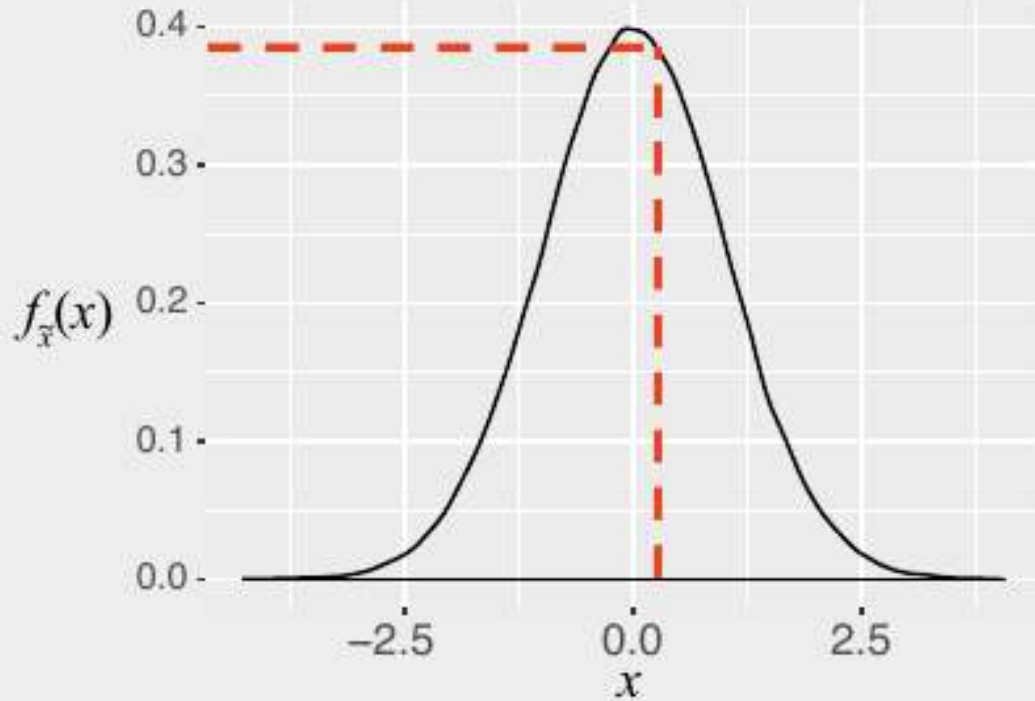
2. Optimization (in general)

BREAK

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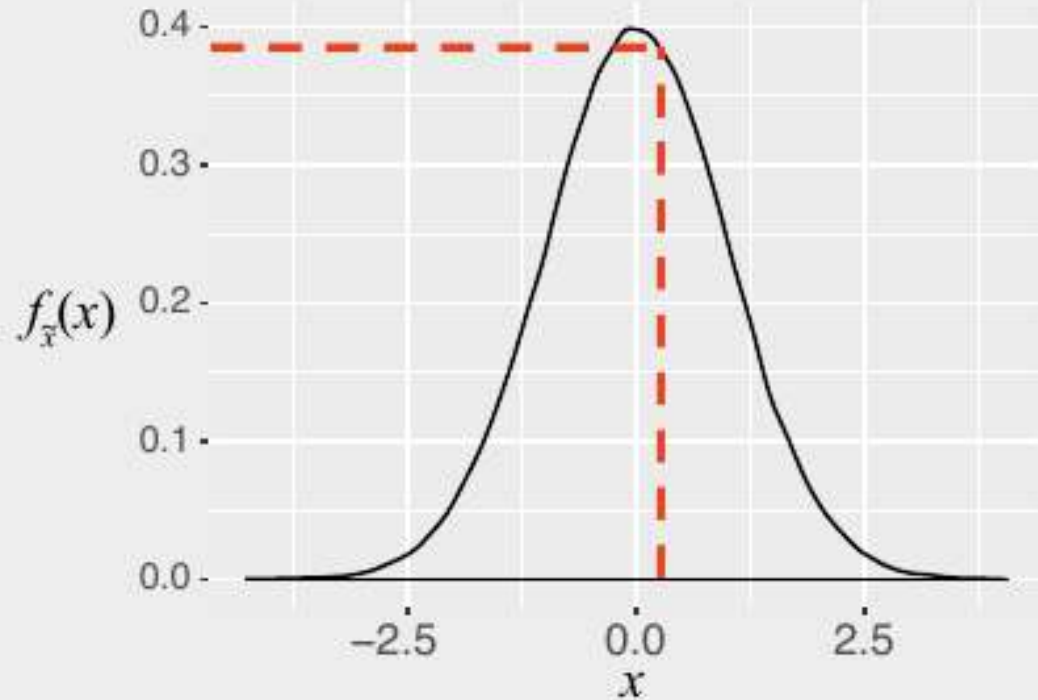
Computing the likelihood



x : an observation

$f(x)$: probability of observing x

Computing the likelihood



x : an observation

$f(x)$: probability of observing x

$\mathcal{L}(\theta|x)$: probability that θ are the true parameters, given that observed x

We want to estimate θ

We actually compute the *log*-likelihood
(converts multiplication to addition)

0.39	0.35	0.24	0.39	0.40	0.11	0.33	0.35	0.07	0.37
------	------	------	------	------	------	------	------	------	------

$$\mathcal{L}(\boldsymbol{\theta}|\mathbf{x}) = f_{\tilde{x}}(x_1) f_{\tilde{x}}(x_2) \dots f_{\tilde{x}}(x_n) = 1.63\text{e-}6$$

$$\log \mathcal{L}(\boldsymbol{\theta}|\mathbf{x}) = f_{\tilde{x}}(x_1) + f_{\tilde{x}}(x_2) + \dots + f_{\tilde{x}}(x_n) = 3$$

Practice Question 1

Observations - Height of students (inches):

```
#> [1] 65 69 66 67 68 72 68 69 63 70
```

- a) Let's say we know that the height of students, $\tilde{x}$, in a classroom follows a normal distribution. A professor obtains the above height measurements students in her classroom. What is the log-likelihood that $\tilde{x} \sim \mathcal{N}(68, 4)$? In other words, compute $\ln \mathcal{L}(\mu = 68, \sigma = 4)$.
- b) Compute the log-likelihood function using the same standard deviation ($\sigma = 4$) but with the following different values for the mean, μ : 66, 67, 68, 69, 70. How do the results compare? Which value for μ produces the highest log-likelihood?

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$f(x)$

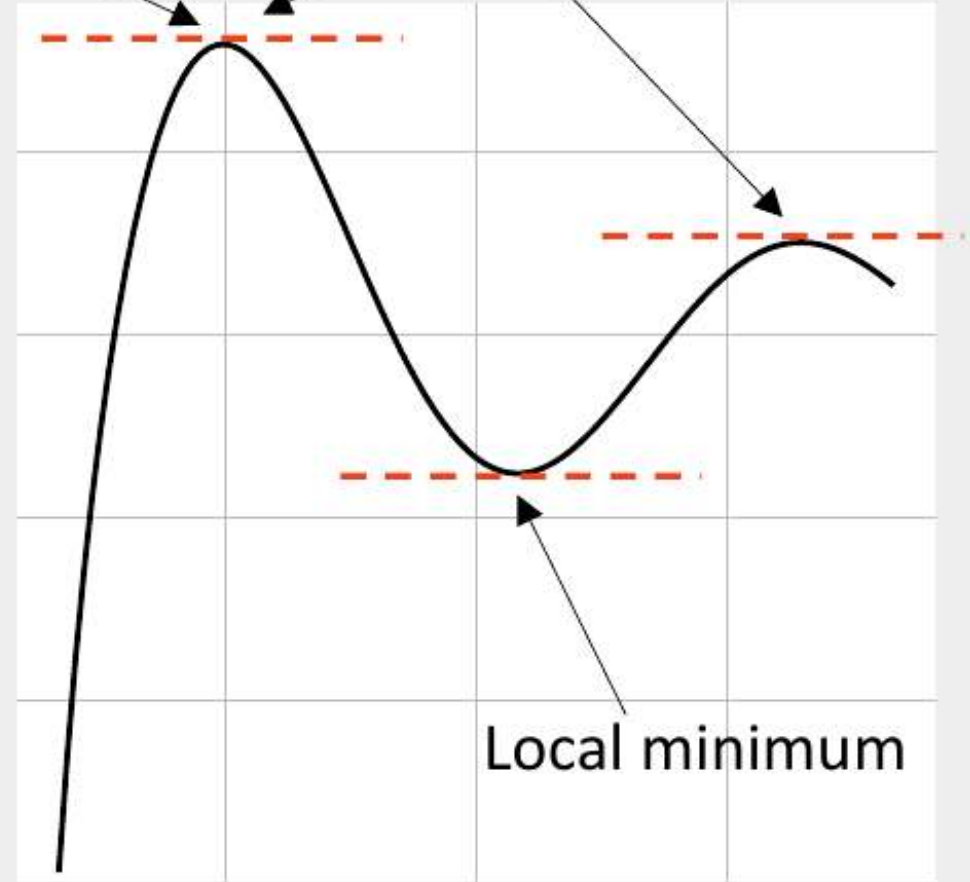
Global maximum

Local maximum

$f(x)$

Local minimum

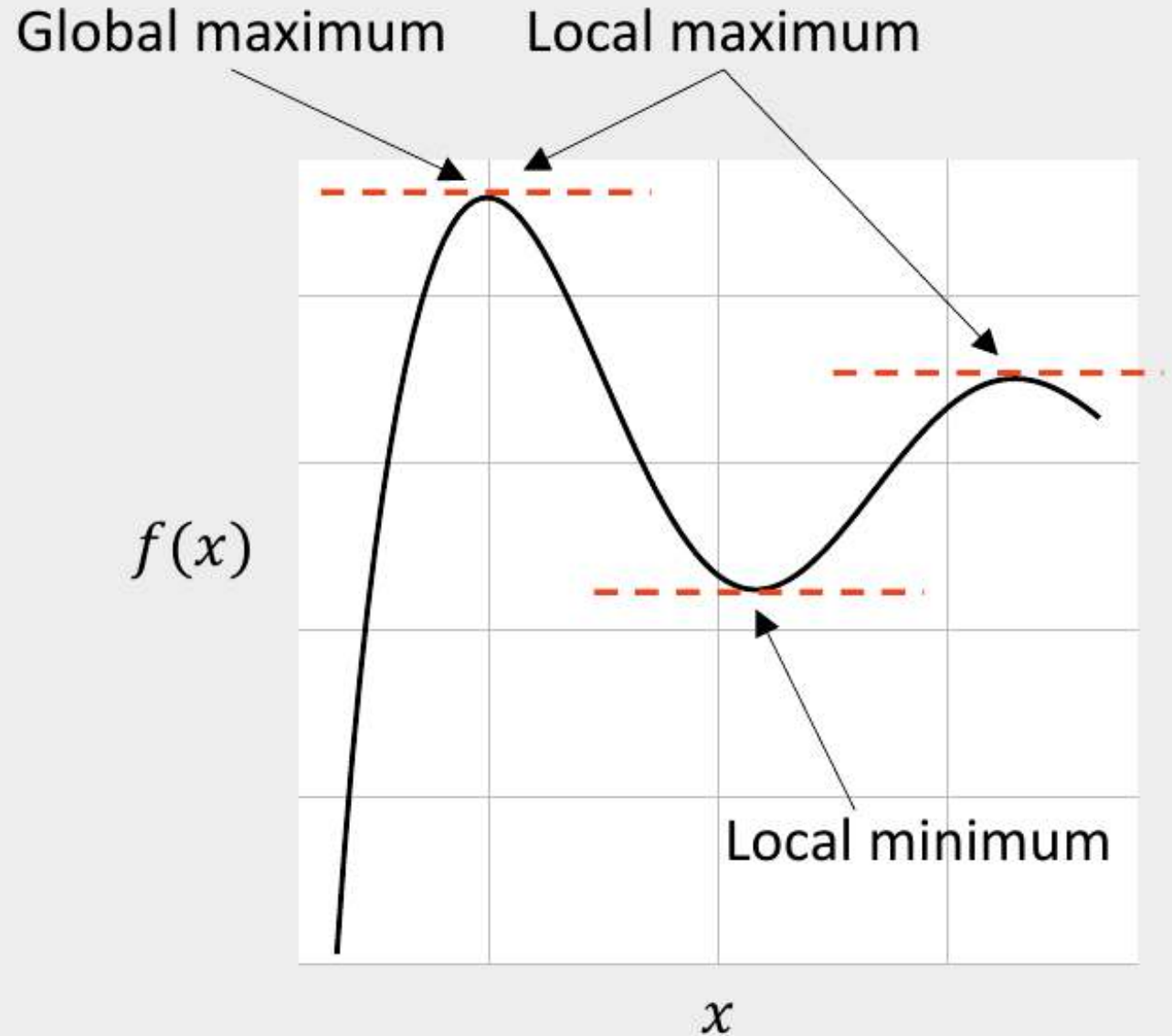
x



First order necessary condition

x^* is a “stationary point” when

$$\frac{df(x^*)}{dx} = 0$$



First order necessary condition

x^* is a “stationary point” when

$$\frac{df(x^*)}{dx} = 0$$

Second order sufficiency condition

x^* is a local *maximum* when

$$\frac{d^2f(x^*)}{dx^2} < 0$$

x^* is a local *minimum* when

$$\frac{d^2f(x^*)}{dx^2} > 0$$

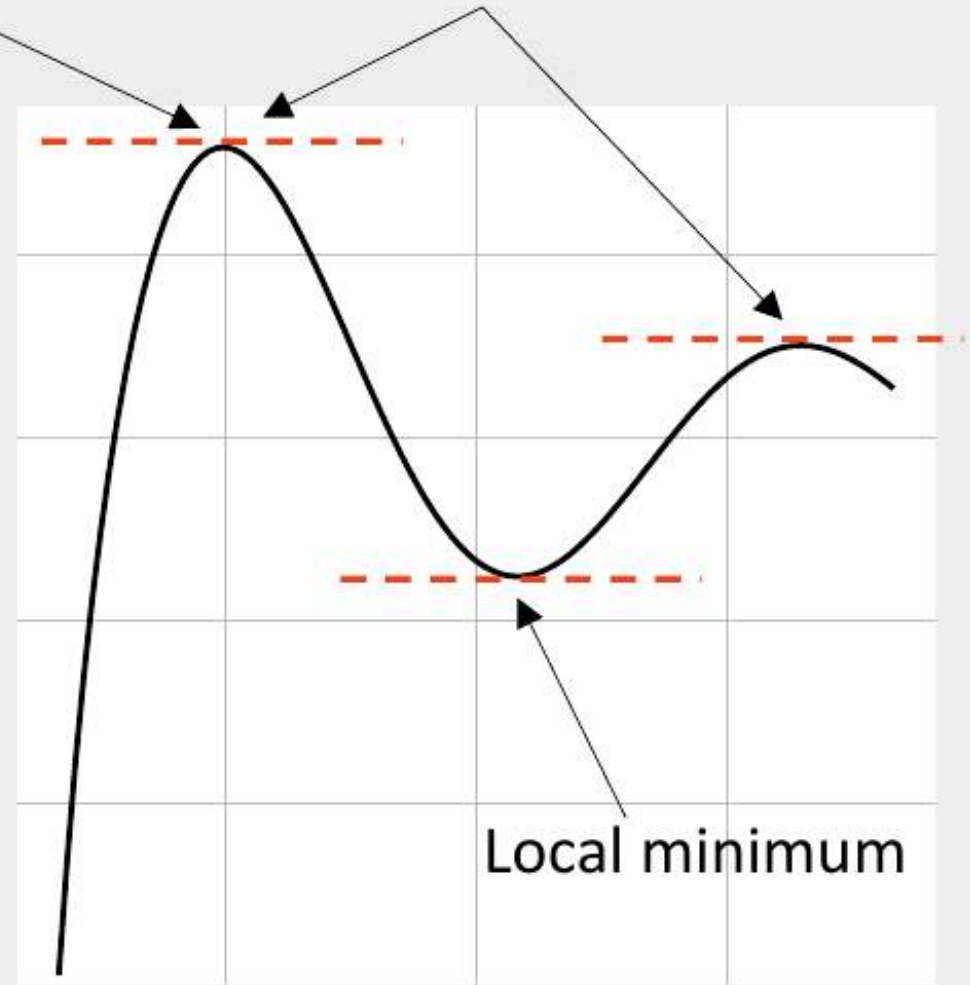
Global maximum

Local maximum

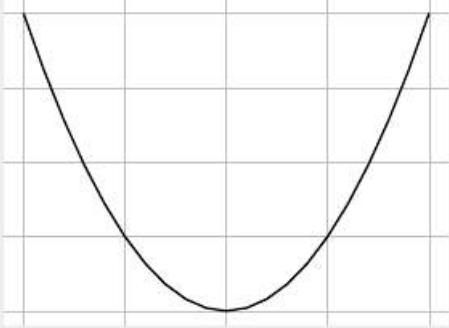
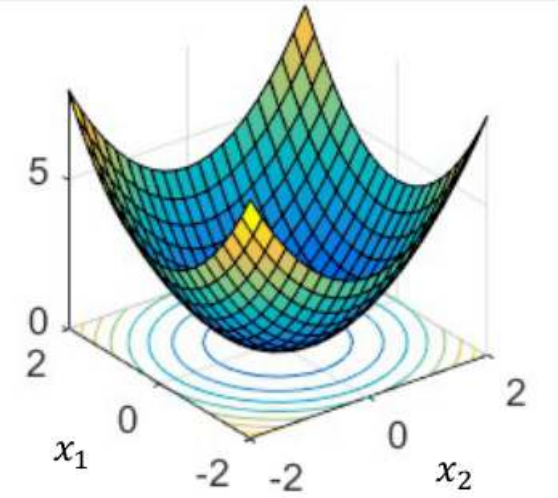
$f(x)$

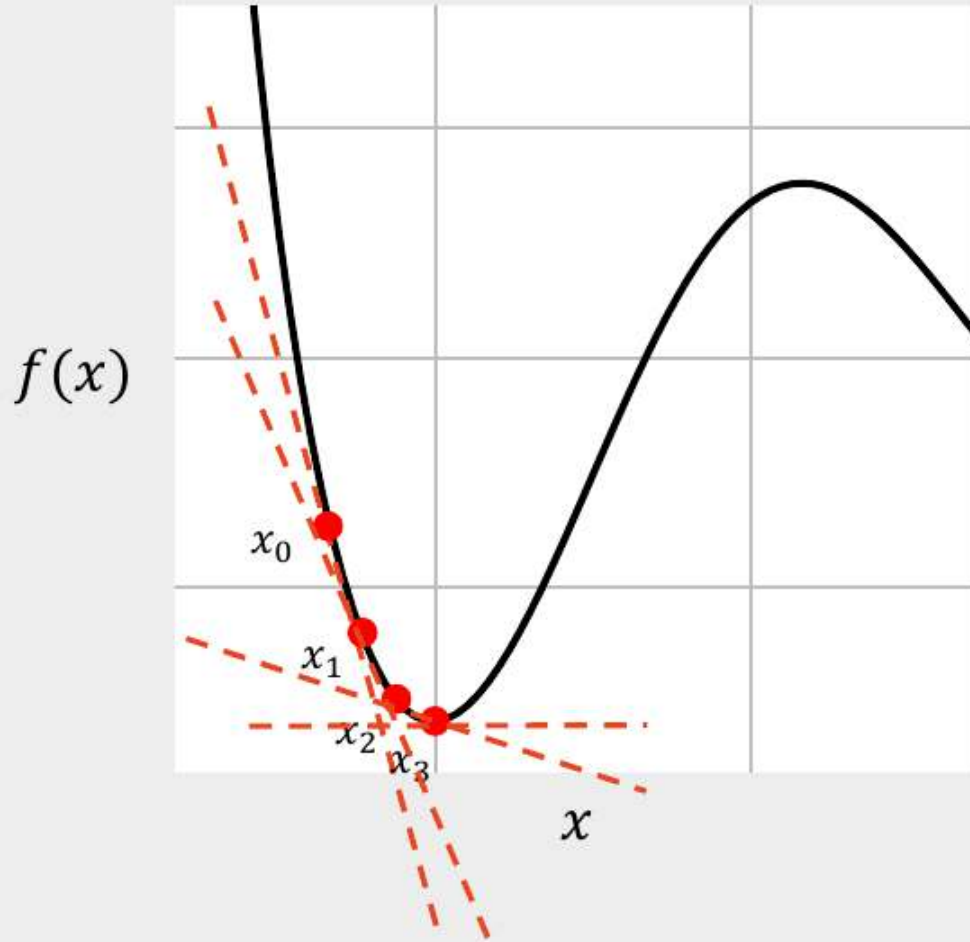
Local minimum

x



Optimality conditions for local **minimum**

Number of dimensions	First order condition	Second order condition	Example
One	$\frac{df(x^*)}{dx} = 0$	$\frac{d^2f(x^*)}{dx^2} > 0$	
Multiple	“Gradient” $\nabla f(x_1, x_2, \dots, x_n)$	“Hessian” $\nabla^2 f(x_1, x_2, \dots, x_n)$	
	$= \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]$ $= [0, 0, \dots, 0]$	$= \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$ Must be “positive definite”	



Gradient Descent Method:

1. Choose a starting point, x_0
2. At that point, compute the gradient, $\nabla f(x_0)$
3. Compute the next point, with a step size γ :

$$x_{n+1} = x_n - \gamma \nabla f(x_n)$$

*Stop when $\nabla f(x_n) < \delta$ Very small number
or

*Stop when $(x_{n+1} - x_n) < \delta$

Practice Question 2

Consider the following function:

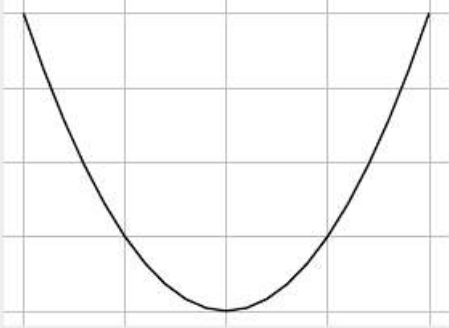
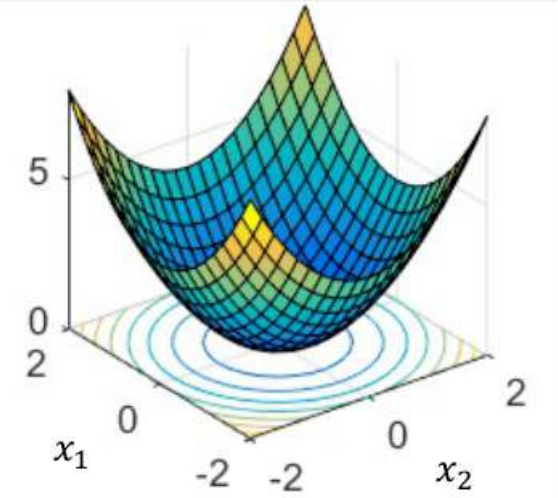
$$f(x) = x^2 - 6x$$

The gradient is:

$$\nabla f(x) = 2x - 6$$

Using the starting point $x = 1$ and the step size $\gamma = 0.3$, apply the gradient descent method to compute the next **three** points in the search algorithm.

Optimality conditions for local **minimum**

Number of dimensions	First order condition	Second order condition	Example
One	$\frac{df(x^*)}{dx} = 0$	$\frac{d^2f(x^*)}{dx^2} > 0$	
Multiple	“Gradient” $\nabla f(x_1, x_2, \dots, x_n)$	“Hessian” $\nabla^2 f(x_1, x_2, \dots, x_n)$	
	$= \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]$ $= [0, 0, \dots, 0]$	$= \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$ Must be “positive definite”	

Practice Question 3

Consider the following function:

$$f(\underline{x}) = x_1^2 + 4x_2^2$$

The gradient is:

$$\nabla f(\underline{x}) = \begin{bmatrix} 2x_1 \\ 8x_2 \end{bmatrix}$$

Using the starting point $\underline{x}_0 = [1, 1]$ and the step size $\gamma = 0.15$, apply the gradient descent method to compute the next **three** points in the search algorithm.

Download the [logitr-cars](#) repo from GitHub

Estimating utility models

1. Open `logitr-cars.Rproj`
2. Open `code/3.1-model-mnl.R`

Maximum likelihood estimation

$$\begin{aligned}\tilde{u}_j &= v_j + \tilde{\varepsilon}_j \\ &= \beta_1 x_{j1} + \beta_2 x_{j2} + \dots + \tilde{\varepsilon}_j \\ &= \boldsymbol{\beta}' \mathbf{x}_j + \tilde{\varepsilon}_j\end{aligned}$$

Estimate $\boldsymbol{\beta} = [\beta_1, \beta_2, \dots, \beta_n]$
by maximizing the likelihood function

$$\begin{aligned}\text{minimize } -\log \mathcal{L} &= - \sum_{j=1}^J P_j (\boldsymbol{\beta} | \mathbf{x})^{y_j} \\ &\text{with respect to } \boldsymbol{\beta}\end{aligned}$$

$y_j = 1$ if alternative j was chosen

$y_j = 0$ if alternative j was not chosen

For logit model:

$$P_j = \frac{e^{v_j}}{\sum_{k=1}^J e^{v_k}} = \frac{e^{\boldsymbol{\beta}' \mathbf{x}_j}}{\sum_{k=1}^J e^{\boldsymbol{\beta}' \mathbf{x}_k}}$$

Break

05 : 00

Week 8: *Optimization & MLE*

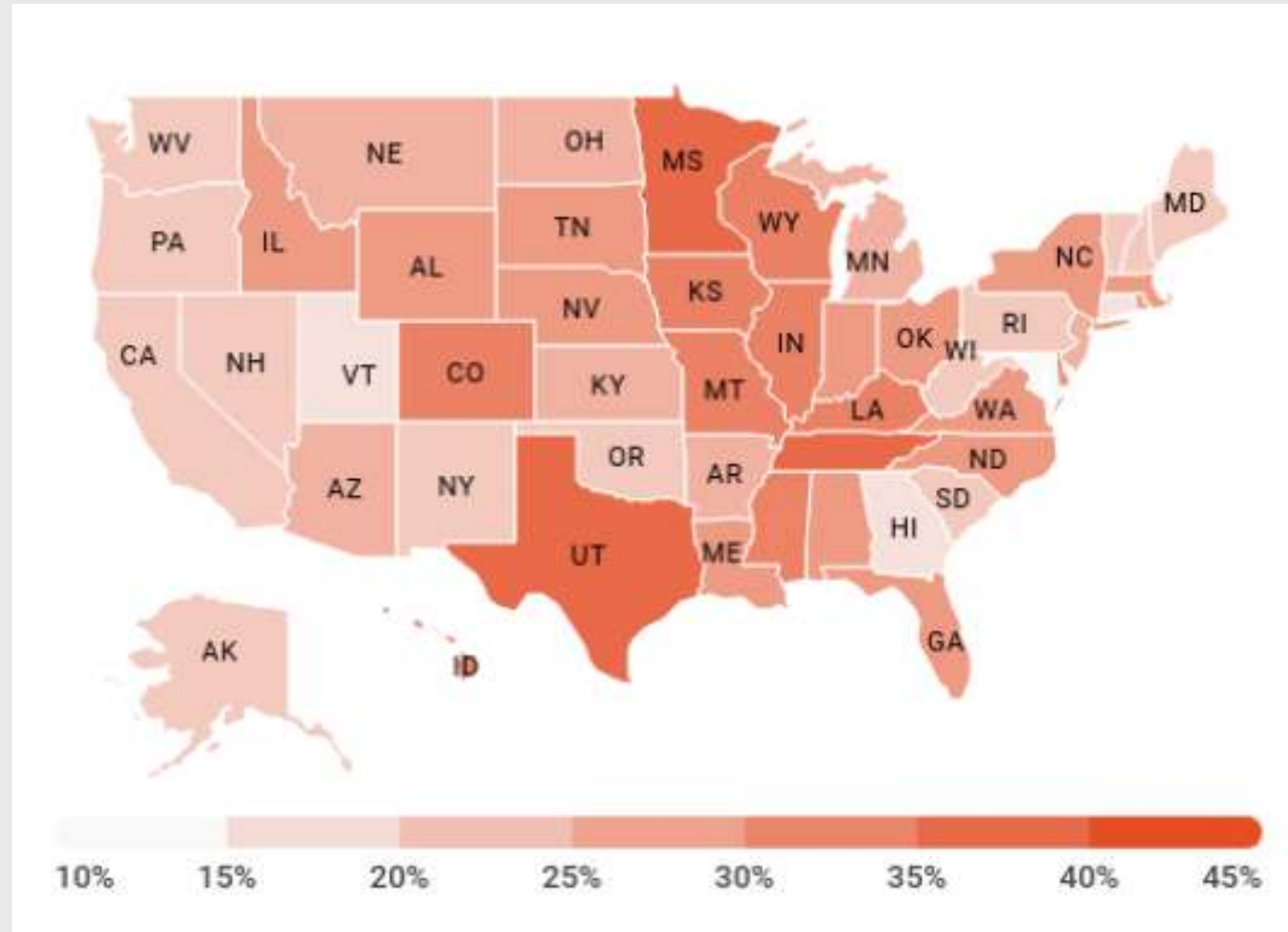
1. Maximum likelihood estimation
2. Optimization (in general)

BREAK

3. **Joins**

4. Pilot data cleaning

What's wrong with this map?



Likely culprit: Merging two columns

```
head(names)
```

```
#>           state_name
#> 1           Alabama
#> 2           Alaska
#> 3           Arizona
#> 4           Arkansas
#> 5 Armed Forces Africa
#> 6 Armed Forces Americas
```

```
head(abbs)
```

```
#> state_abb
#> 1      AA
#> 2      AE
#> 3      AE
#> 4      AE
#> 5      AE
#> 6      AK
```

```
result <- cbind(names, abbs)
head(result)
```

```
#>           state_name state_abb
#> 1           Alabama      AA
#> 2           Alaska      AE
#> 3           Arizona      AE
#> 4           Arkansas      AE
#> 5 Armed Forces Africa      AE
#> 6 Armed Forces Americas      AK
```

Joins

1. `inner_join()`
2. `left_join()` / `right_join()`
3. `full_join()`

Example: `band_members` & `band_instruments`

`band_members`

```
#> # A tibble: 3 × 2
#>   name  band
#>   <chr> <chr>
#> 1 Mick  Stones
#> 2 John  Beatles
#> 3 Paul  Beatles
```

`band_instruments`

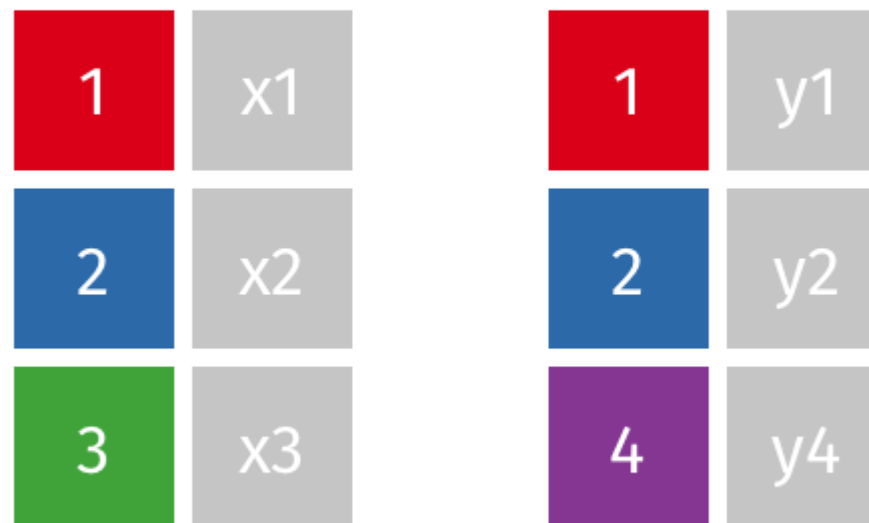
```
#> # A tibble: 3 × 2
#>   name  plays
#>   <chr> <chr>
#> 1 John  guitar
#> 2 Paul  bass
#> 3 Keith guitar
```

inner_join()

```
band_members %>%  
  inner_join(band_instruments)
```

```
#> # A tibble: 2 × 3  
#>   name  band  plays  
#>   <chr> <chr> <chr>  
#> 1 John  Beatles guitar  
#> 2 Paul  Beatles  bass
```

inner_join(x, y)

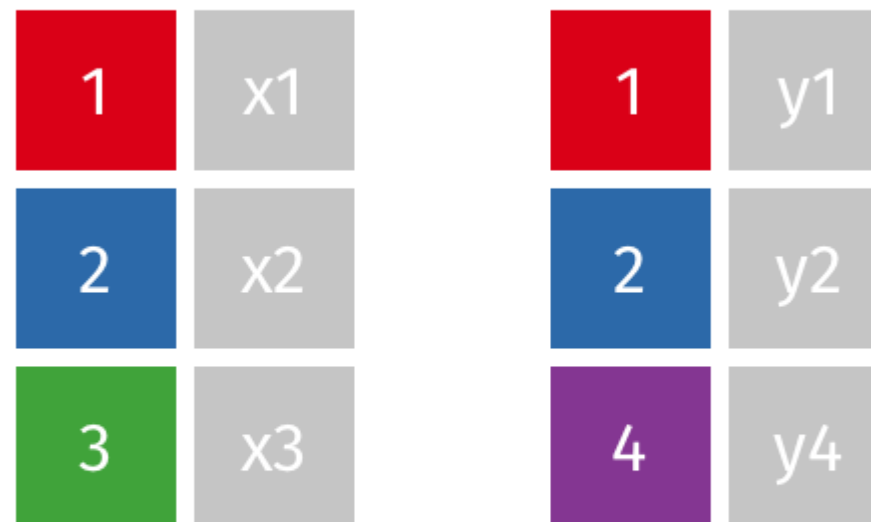


full_join()

```
band_members %>%  
  full_join(band_instruments)
```

```
#> # A tibble: 4 × 3  
#>   name band    plays  
#>   <chr> <chr>   <chr>  
#> 1 Mick  Stones  <NA>  
#> 2 John  Beatles guitar  
#> 3 Paul  Beatles bass  
#> 4 Keith <NA>    guitar
```

full_join(x, y)



left_join()

```
band_members %>%  
  left_join(band_instruments)
```

```
#> # A tibble: 3 × 3  
#>   name  band  plays  
#>   <chr> <chr> <chr>  
#> 1 Mick  Stones <NA>  
#> 2 John  Beatles guitar  
#> 3 Paul  Beatles bass
```

left_join(x, y)

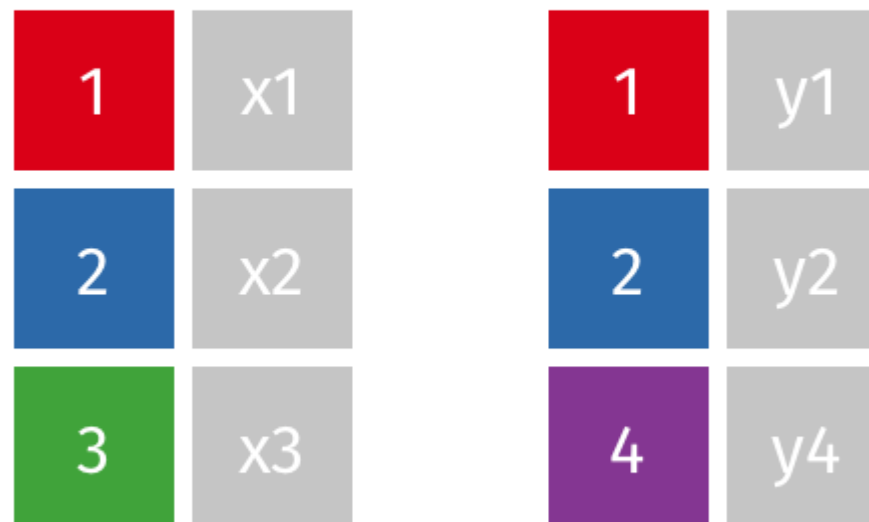
1	x1	1	y1
2	x2	2	y2
3	x3	4	y4

right_join()

```
band_members %>%  
  right_join(band_instruments)
```

```
#> # A tibble: 3 × 3  
#>   name  band  plays  
#>   <chr> <chr> <chr>  
#> 1 John  Beatles guitar  
#> 2 Paul  Beatles bass  
#> 3 Keith <NA>   guitar
```

right_join(x, y)



Specify the joining variable name

```
band_members %>%  
  left_join(band_instruments)
```

```
#> Joining with `by = join_by(name)`
```

```
#> # A tibble: 3 × 3  
#>   name band plays  
#>   <chr> <chr> <chr>  
#> 1 Mick  Stones <NA>  
#> 2 John  Beatles guitar  
#> 3 Paul  Beatles bass
```

```
band_members %>%  
  left_join(band_instruments,  
            by = 'name')
```

```
#> # A tibble: 3 × 3  
#>   name band plays  
#>   <chr> <chr> <chr>  
#> 1 Mick  Stones <NA>  
#> 2 John  Beatles guitar  
#> 3 Paul  Beatles bass
```

Specify the joining variable name

If the names differ, use `by = c("left_name" = "joining_name")`

```
band_members
```

```
#> # A tibble: 3 × 2  
#>   name band  
#>   <chr> <chr>  
#> 1 Mick  Stones  
#> 2 John  Beatles  
#> 3 Paul  Beatles
```

```
band_instruments2
```

```
#> # A tibble: 3 × 2  
#>   artist plays  
#>   <chr>   <chr>  
#> 1 John   guitar  
#> 2 Paul   bass  
#> 3 Keith  guitar
```

```
band_members %>%  
  left_join(band_instruments2,  
            by = c("name" = "artist"))
```

```
#> # A tibble: 3 × 3  
#>   name band plays  
#>   <chr> <chr> <chr>  
#> 1 Mick  Stones <NA>  
#> 2 John  Beatles guitar  
#> 3 Paul  Beatles bass
```

Specify the joining variable name

Or just rename the joining variable in a pipe

```
band_members
```

```
#> # A tibble: 3 × 2  
#>   name band  
#>   <chr> <chr>  
#> 1 Mick  Stones  
#> 2 John  Beatles  
#> 3 Paul  Beatles
```

```
band_instruments2
```

```
#> # A tibble: 3 × 2  
#>   artist plays  
#>   <chr>   <chr>  
#> 1 John   guitar  
#> 2 Paul   bass  
#> 3 Keith  guitar
```

```
band_members %>%  
  rename(artist = name) %>%  
  left_join(band_instruments2,  
            by = "artist")
```

```
#> # A tibble: 3 × 3  
#>   artist band    plays  
#>   <chr>   <chr>   <chr>  
#> 1 Mick   Stones  <NA>  
#> 2 John   Beatles guitar  
#> 3 Paul   Beatles bass
```

Your turn

15:00

1) Create a new data frame called `state_data` by joining the `state_abbs` and `state_regions` data frames. The result should be a data frame with variables `state_name`, `state_abb`, and `state_region`. It should look like this:

```
head(state_data)
```

```
#> # A tibble: 6 × 3
#>   state_name state_abb state_region
#>   <chr>      <chr>      <chr>
#> 1 Alabama    AL          Southeast
#> 2 Alaska     AK          Pacific
#> 3 Arizona    AZ          Mountain
#> 4 Arkansas   AR          Delta States
#> 5 California CA          Pacific
#> 6 Colorado   CO          Mountain
```

2) Join the `state_data` data frame to the `wildlife_impacts` data frame, adding the variables `state_region` and `state_name`.

```
glimpse(wildlife_impacts)
```

```
#> Rows: 56,978
#> Columns: 23
#> $ state_abb      <chr> "FL", "IN", NA, NA, NA, "FL", "FL", NA, NA, "FL",
#> $ state_name     <chr> "Florida", "Indiana", NA, NA, NA, "Florida", "Flo
#> $ state_region   <chr> "Southeast", "Corn Belt", NA, NA, NA, "Southeast",
#> $ incident_date  <dbl> 2018-12-31, 2018-12-29, 2018-12-29, 2018-12-27,
#> $ airport_id     <chr> "KMIA", "KIND", "ZZZZ", "ZZZZ", "ZZZZ", "KMIA", "
#> $ airport        <chr> "MIAMI INTL", "INDIANAPOLIS INTL ARPT", "UNKNOWN",
#> $ operator       <chr> "AMERICAN AIRLINES", "AMERICAN AIRLINES", "AMERIC
#> $ atype          <chr> "B-737-800", "B-737-800", "UNKNOWN", "B-737-900",
#> $ type_eng       <chr> "D", "D", NA, "D", "D", "D", "D", "D", "D", "D",
#> $ species_id     <chr> "UNKBL", "R", "R2004", "N5205", "J2139", "UNKB",
#> $ species        <chr> "Unknown bird - large", "Owls", "Short-eared owl",
#> $ damage         <chr> "M?", "N", NA, "M?", "M?", "N", "N", "N", "N", "N",
#> $ num_engs       <dbl> 2, 2, NA, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,
#> $ incident_month <dbl> 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12,
#> $ incident_year  <dbl> 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2018,
#> $ time_of_day    <chr> "Day", "Night", NA, NA, NA, "Day", "Night", NA, NA,
#> $ time           <dbl> 1207, 2355, NA, NA, NA, 955, 948, NA, NA, 1321,
#> $ height         <dbl> 700, 0, NA, NA, NA, NA, 600, NA, NA, 0, NA, 0, NA,
#> $ speed          <dbl> 200, NA, NA, NA, NA, NA, 145, NA, NA, 130, NA, NA,
#> $ phase_of_flt   <chr> "Climb", "Landing Roll", NA, NA, NA, "Approach",
#> $ sky            <chr> "Some Cloud", NA, NA, NA, NA, NA, "Some Cloud", NA,
#> $ precip         <chr> "None", NA, NA, NA, NA, NA, "None", NA, NA, "None",
#> $ cost_repairs_infl_adj <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,
```

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1. Maximum likelihood estimation

2. Optimization (in general)

BREAK

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Download the [conjoint_buttons](#) template repo

Cleaning surveydown survey data

1. Open `survey.Rproj`
2. Open `code/3_data_cleaning.R`

Team time

For the rest of class, work with your team to finalize your pilot surveys and get them live!